Journal of Further and Higher Education

ISSN: 0309-877X (Print) 1469-9486 (Online) Journal homepage: www.tandfonline.com/journals/cjfh20

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To cite this article: Sara de Freitas, Guy J. Curtis & Stephen M. Ritchie (2025) A multi-layered evaluation of university-wide education interventions to improve student engagement and retention, Journal of Further and Higher Education, 49:10, 1348-1362, DOI: 10.1080/0309877X.2025.2560999

To link to this article: https://doi.org/10.1080/0309877X.2025.2560999

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Published online: 18 Sep 2025.

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JOURNAL OF FURTHER AND HIGHER EDUCATION
2025, VOL. 49, NO. 10, 1348–1362
https://doi.org/10.1080/0309877X.2025.2560999

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A multi-layered evaluation of university-wide education interventions to improve student engagement and retention

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ABSTRACT

The strategic benefit of student retention is financial for institutions, economic for states and personal for individual students. Nonetheless, enacting and evaluating effective innovations to improve retention is challenging. This paper reports a multi-layered set of methods combining a quasi-experimental design, statistical methods with a dynamic learning analytic approach based upon baseline of data, multiple data sets and mixed methods (quantitative and qualitative) to explore how student engagement, progression and retention were affected by learning interventions of different varieties deployed across a university. With a cross-university team including executive and faculty leadership, our strategy-guided intervention reached 65% of the university's first-year students (with 2,990 students from 36 modules/units, and 24 disciplines, involved in the study). Our main findings show that the most benefit in retention was gained by the use of supplemental instruction, which coincided with increased satisfaction of students' psychological needs as a proxy measure of engagement. Cost-benefit modelling indicated that our strategic interventions also led to substantial revenue gains. This paper provides a framework for higher educational institutions to evaluate similar programmes and demonstrates the efficacy of the programme that we assessed.

ARTICLE HISTORY
Received 5 September 2023
Accepted 9 September 2025

KEYWORDS
Higher education; learning and teaching; student retention; student engagement; learning interventions; learning gain

The retention – or progression – of university students has been extensively covered in existing research literature, and the key findings have consistently shown that student attrition is a major concern for many global higher-education institutions (e.g. Aulck et al. 2016; Daempfle 2003; Pedler, Willis, and Nieuwoudt 2022). This concern is especially prevalent when students are from disadvantaged socio-economic groups, e.g. 'first-in-family', diverse/indigenous, mature-age, and/or external status (Devlin et al., 2012; Walpole 2003; Yorke and Thomas 2003). Although there are complexities with how the data are collected, the Organisation for Economic Co-operation and Development (2016) reports

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completion rates from a longitudinal study of cohorts from Bachelor's degrees or equivalent in 2014 (see A9.1 table on p. 166). These data indicate completion rates of over 80% for tertiary education in the United Kingdom, Denmark and New Zealand at the upper end, with Austria, Sweden and Estonia at the lower end, with only 50–60%. Overall, the report states that 'the average completion rate among countries with available data is 49% within the theoretical duration and 68% three years later' (OECD 2016, 168). This means that, on average, among the OECD countries, routinely around a third of the students are discontinuing their degrees, despite quite high participation rates.

At an institutional level, progression rates are concerning, given that non-completing students represent a source of lost revenue (DeShields, Kara, and Kaynak 2005). From a societal perspective, attrition may result in poor outcomes for students (e.g. lowered self-efficacy and career prospects; Crosling, Heagney, and Thomas 2009). However, despite the importance of optimising student progression, there are significant gaps in the literature currently; for example, we were unable to find convincing progression benchmarking data for comparative studies. We also found that most studies that focused upon innovative targeted interventions were either single-discipline focused (e.g. Terry 2001), lacked clear methodologies, used varied and unclear definitions of retention/progression and/or often included small sample sizes (e.g. Metzner and Bean 1987).

To address this, a study design would need to link the existing 'true-cohort' (a cohort using actual rather than projected data), longitudinal student retention data, with a range of learning-focused interventions that could positively impact students' experiences (e.g. greater engagement). It would also need to integrate large student datasets, with student-engagement survey data showing the efficacy of university-wide interventions. Yet, this kind of study has not been undertaken previously. To undertake a 'multi-layered' evaluation of a suite of innovative university-wide educational and teaching interventions at a case-study site, which borrows methods from psychological and data analytic perspectives, our evaluation team included two senior-executives (data-science and education backgrounds), academic and professional staff members, two research students (psychology), the business-intelligence, a student-advisor network and learning/teaching units.

Another challenge we faced in the study design was that we still have relatively little literature about the impact of teaching and learning interventions upon retention and its causes (Eather et al. 2022). From Eather et al.'s (2022) systematic review of programmes targeting student satisfaction and retention, we see that many interventions are not in fact focused on learning and teaching; studies have mostly examined single interventions or single outcomes; or programmes and their evaluations have been relatively small in scale.

However, from the literature that has been published; for example, we see from Tinto (1975, 1987) that staff and student interaction as well as students' academic and social integration at university are significant influencers upon progression. A more recent study surveying 2,451 students (representing 40 universities and at least 15 disciplines) found that a lack of social 'esteem' (with university peers/staff), alongside low levels of academic success, predict higher withdrawal intentions and burnout levels (which are often considered to be the opposite/reverse of engagement; C. J. Williams, Dziurawiec, and Heritage 2018). Similarly, others have noted that students' engagement and

competence/achievement within their studies are critical factors (e.g. Hilts, Part, and Bernacki 2018; Kuh et al. 2008). From the literature, it appears that a range of study-based characteristics influence rates of student attrition.

Alongside demographic and study-based characteristics, high-quality teaching is also believed to be particularly influential upon student engagement and progression (Stronge 2018). Furthermore, high-quality teaching may be particularly valuable in the earlier stages of a degree, as Kift, Nelson, and Clarke (2010) found that the dropout rate was exacerbated in the first year of university studies. With this in mind, Crosling, Heagney, and Thomas (2009) provided one of the few studies that focused upon student progression from a teaching/learning perspective. Here, the scholars found a close connection between quality-assurance and improved student progression in Australian higher education. Additionally, de Freitas et al. (2015) adopted a longitudinal-study examining the causes of attrition rates in five years of student-data (51,182 students at one university, with 61 million data elements). They found that some factors regarding learning/teaching were key attrition risk factors (e.g. unhappiness with performance; de Freitas et al. 2015). The literature and interest in measuring 'learning gain' has become a subject of interest recently with early studies adopting pre- and post-study models (Colt et al. 2011), small sample-sizes (Kooloos et al. 2011) and mixed-methods (Mathabathe and Potgieter 2014).

The literature suggests that supplemental instruction in the form of peer mentoring or peer-led study is particularly effective in promoting student learning, satisfaction and retention (Eather et al. 2022). One reason for the efficacy of such programmes may be that they help to satisfy students' need for belonging (Pedler, Willis, and Nieuwoudt 2022). More broadly, self-determination theory suggests that increasing satisfaction (and reducing frustration) in personal relationships, self-efficacy and independence promotes motivation, well-being and, ultimately, may promote student success and retention (Altonji, Banos, and Harada 2019; Longo et al. 2016; Pedler, Willis, and Nieuwoudt 2022; Ryan and Deci 2000)

The present study

From reviewing the literature it is evident that learning and teaching factors, if innovatively optimised, can improve progression. In our study, interventions were enacted in first-year units/modules from eight schools (24 disciplines) at one university. Many of the selected interventions were focussed on improving student engagement (e.g. active learning). For our study, we were concerned with testing a diverse range of learning interventions, such as student mentoring, peer-assisted learning (Dawson et al. 2014), welcome-videos, automated-assessment, gamification and educational software.

This evaluation brings together the longitudinal data of 'true-cohorts' – focusing on correlated data for student performance, continuation and progression – with a study that aims to measure student engagement through the use of a validated survey instrument based on self-determination theory (SDT; Ryan and Deci 2000). A summary of the financial outcomes to support senior-executives who want to implement these kinds of teaching interventions in their universities is also reported. Together, the pedagogic, student outcome and financial benefits make

this a more systematic approach to examining progression gains. Our research questions were:

- What is the impact of in-term interventions on student engagement and progression?
- What is the cost-benefit of these interventions?

Methods and measures

A pragmatic approach was used to evaluate the efficacy of an initiative to improve student engagement and progression at a small–medium sized Australian university given Reichardt and Cook's (1979) advice: 'Evaluators would be wise to use whatever methods are best suited to their research needs, regardless of the methods' traditional affiliations' (p. 19). At the time of the study, the university had ~ 23,000 students (59.0% female, 5.7% of non-English speaking background, 10.2% with a disability, and 19.7% from a low socio-economic status household), with ~ 4,600 in their first-year.

We studied two concurrent progression intervention initiatives, a diverse range of interventions named the School-Led Teaching Effectiveness Programme (STEP) and Supplemental Instruction (SI). The SI programme has been deployed in over 1,500 tertiary education institutions from 29 countries (Martin 2008). SI interventions are based on a simple idea: 'find the trouble spots in the curriculum; and with the permission of the lecturer, ask a student who has done well in the subject to help others master the content through small group collaborative activities' (Martin 2008, 3). The SI model we employed was 'peer-assisted study sessions (PASS)' (Dawson et al. 2014; Mellor et al. 2015).

The multi-layered evaluation was conducted in three phases. The first phase of the evaluation involved measuring changes in student engagement over ten weeks during the semester in which STEP and SI were implemented. The second phase analysed longitudinal data on student enrolments for units in which STEP or SI featured. The final phase of the evaluation set out to determine the cost-benefit of investing resources into STEP and SI. This phase was conducted post hoc.

STEP and SI implementation

To gain maximum impact for students in STEP and SI units, we focused upon the university's large first-year units, and those with historically-low progression rates. Half of the interventions were SI and the remaining 50%, which included the diversity of interventions noted below, were called STEP. We analysed SI and STEP interventions separately and in aggregate to better understand their impact on engagement and retention. We also focused interventions in two Schools with the lowest student-satisfaction ratings. The STEP interventions ranged from unit welcome-videos (teaching staff providing students with video unit overviews), a redevelopment of the existing student-mentoring programme, gamification using progress bars and badging, automated online science labs, Learning Management System (LMS) support for improving sites, online forum development, university-wide introduction of software

to improve students' competence in writing skills and assignment calculators that allow students to plan for their assessments (providing students with progressive weighted unit marks after each assessment item).

Reach of the STEP and SI interventions

Within this study, 65% of the first-year students ($n = 2{,}990$) were reached through the intervention programmes in 2016. Overall, 37 units were included in the study across all of the university's 24 discipline areas.

Phase 1 method: measuring student engagement

Self-determination theory's (SDT) mini-theory of 'basic needs' underpinned the student engagement analysis. Satisfaction of students' need for competence, autonomy and relatedness align with educational engagement (Reeve 2012), motivation (Ryan and Deci 2000) and performance (Schneider and Preckel 2017). According to SDT (Ryan and Deci 2000), if core psychological needs are met, students will remain satisfied and intrinsically motivated (thereby increasing progression rates; Adarlo, Wang, and King 2025; Reeve 2012). Conversely, need frustration undermines well-being (Tindall and Curtis 2019) and may therefore be an important contributor to student attrition. Thus, our study posited that, by enhancing social and academic experiences, both engagement and progression will increase via increases to students' need satisfaction and decreases in their need frustration.

Based upon SDT, the Need Satisfaction and Frustration Scale (NSFS; Longo et al. 2016) is the validated instrument that we used to measure the satisfaction and frustration of students' basic psychological needs (autonomy, relatedness and competence). Satisfaction of these psychological needs is related to well-being, engagement and motivation, while frustration of these needs is related to stress, disengagement and burnout (Longo et al. 2016;). The NSFS has recently been used to assess the impact of changes to campus-wide curriculum and teaching delivery on students' psychological needs (see de Freitas et al. 2021). The NSFS contains 18 items each rated on a 7-point scale anchored with 'strongly agree' and 'strongly disagree'. The NSFS has six 3-item subscales, with three subscales measuring the extent to which students feel their studies provide satisfaction of their basic psychological needs for autonomy, relatedness and competence (e.g. 'In my studies I feel free to decide what to do') and the other three subscales measure need frustration in these domains (e.g. 'In my studies, sometimes I feel a bit rejected by others').

The NSFS was selected for use in our study because it is specifically designed to assess basic psychological needs in an educational context, whereas many other SDT-based measures are targeted to general or workplace need satisfaction. In addition, the NSFS captures need frustration, which is neglected in measures that focus only on the presence or absence of need satisfaction. Moreover, the NSFS has demonstrated excellent psychometric properties in several cross-national studies (e.g. Longo et al. 2016; Longo, Alcaraz-Ibáñez, and Sicilia 2018).

The NSFS was made available to all students within 37 units, the units including SI and STEP interventions as well as the ten largest first-year units that were not using SI or STEP interventions, which formed a quasi-experimental control group. The NSFS was made available to students online early in the semester (time-one: weeks 2–4) and

later in the semester (time-two: weeks 12–14). A prize draw for a gift voucher was used to encourage participation in the survey. There were 3,508 responses to the survey in total. Of these responses, 60 had more than two missing values to NSFS items and nine answered the same for every item (e.g. 'strongly agree' on all items, which is a questionable response indicating inattention to the items). In sum, these 69 substantially incomplete or questionable response sets were removed, leaving 3,439 useable surveys (Time-one *N*s = 401 SI unit students, 394 STEP unit students and 554 comparison unit students; Time-two *N*s = 701 SI unit students, 423 STEP unit students and 966 comparison unit students).

For time-one, students were directed to the NSFS via Learning Management System announcements from their lecturers and via direct email invitations. For time-two, the NSFS was included in the online student evaluation platform along with the university's standard teaching and class evaluation surveys. Because students more typically engage with regular teaching evaluation surveys than one-off surveys, the response rates were higher at time-two, ~25%, than at time-one, ~15%. Ethical approval was obtained to analyse these survey data and other student demographics and analytics.

Phase 2 method: dynamic learning analytics

There were two parts to this phase of the evaluation. First, we explored the relationship between learning gains participating students made and several other data sources to which we had access using a dynamic model for data analysis. In this approach, hypotheses were generated on-the-fly as analyses were conducted to benchmark individual and cohort changes using a comparative range of different data, e.g. demographic, attendance, Cardax (mobility on campus), performance (GPA), continuation and progression, and learning management system (LMS) usage data. In addition to the NSFS, we collected student enrolment data (enrolment in a subsequent semester of study vs no enrolment in the next semester) and information about students' demographics and educational attainment (grades) from the university's student records systems. This method had proved successful in previous studies of student attrition (e.g. de Freitas et al. 2015). However, these analyses produced promising, but complicated, connections, which were challenging to summarise given the large number of students and variables available to us. Because of this, the second phase of analysis was to employ a powerful omnibus exploratory analysis technique.

We adopted Chi-square Automatic Interaction Detection (CHAID) modelling classification tree analysis to help us understand the relationship between student engagement and their progression, i.e. to investigate the potential differences between students who did and did not enrol in units in the next semester after the study STEP/SI interventions. CHAID is a useful approach for identifying the most significant relations between many predictors and one categorical dependent variable, while employing conservative non-parametric statistical assumptions of significance (Kass, 1980). CHAID employs a non-linear iterative process of identifying the most significant multinomial splits of a group of predictor variables using a Bonferroni controlled Chi-squared statistic (Magidson, 1994). Ultimately, CHAID is particularly useful in potentially identifying hidden predictors of group membership that may be practically useful but are not initially theoretically expected or examined in *a priori* planned analyses.

Phase 3 method: financial modelling

As an important part of the situational context, or institutional motivation for investing in any innovative programme of interventions, in addition to the educational considerations, it is critical to consider the cost-benefits of introducing such programmes. To bring this into the study, we adopted a basic financial cost-benefit analysis model that connected the student fees with progression.

Results

Results of time-one and time-two NSFS comparisons

Need Satisfaction and Frustration Scale (NSFS) scores were first checked for normality (skew and kurtosis) and outliers. The data were normally distributed and without outliers.

The same student groups were polled twice during the term (semester). There was some overlap among students who were in units with STEP or SI interventions. Because of this, we separated the students for analysis into groups of those who had been in STEP-only, SI-only, both, or neither (i.e. the comparison units). Table 1 shows that, overall, the time of surveying students (i.e. time-one as compared to time-two) made a significant difference to the reported levels of need satisfaction/frustration. On average, students who were surveyed early in the semester reported significantly lower relatedness satisfaction and competence satisfaction, as well as significantly higher autonomy frustration, relatedness frustration and competence frustration, than students who were surveyed later in the semester. However, while these improvements are encouraging (especially given that student motivation/engagement often decreases throughout the academic semester; Zusho, Pintrich, and Coppola 2003), they do not speak to any specific difference between SI, STEP and the comparison units.

As can be seen in Table 1, for the most part, regardless of the intervention programme, students' basic psychological needs were, on average, more satisfied and less frustrated at the later compared to the earlier time of surveying. Regardless of intervention, students' average relatedness and competence satisfaction were higher at time-two than at time-one, and their competence frustration was lower

Table 1. Means and standard deviations of NSFS scores at time-one (T1) and time-two (T2).

	STEP *M* (SD)	SI *M* (SD)	Combined STEP and SI *M* (SD)	Comparison Units *M* (SD)	Overall *M* (SD)
Autonomy satisfaction T1	5.26 (1.20)	5.25 (1.16)	5.25 (1.15)	5.24 (1.16)	5.25 (1.16)
Autonomy satisfaction T2	5.19 (1.22)	5.27 (1.17)	5.25 (1.18)	5.15 (1.22)	5.20 (1.20)
Autonomy frustration T1	3.69 (1.31)	3.57 (1.27)	3.63 (1.28)	3.64 (1.25)	3.63 (1.27)
Autonomy frustration T2	**3.75 (1.36)**	**3.70 (1.35)**	3.70 (1.35)	**3.77 (1.37)**	**3.73 (1.36)**
Relatedness satisfaction T1	4.48 (1.13)	4.53 (1.18)	4.55 (1.14)	4.42 (1.13)	4.50 (1.14)
Relatedness satisfaction T2	**4.77 (1.18)**	**4.82_a (1.20)**	**4.77 (1.19)**	**4.68_b (1.16)**	**4.73 (1.18)**
Relatedness frustration T1	3.45 (1.29)	3.38 (1.26)	3.39 (1.28)	3.37 (1.29)	3.39 (1.28)
Relatedness frustration T2	**3.30 (1.41)**	3.30 (1.40)	3.29 (1.42)	3.27 (1.38)	**3.28 (1.40)**
Competence satisfaction T1	4.41_b (1.19)	4.60_a (1.16)	4.52_a (1.16)	4.52_a (1.19)	4.52 (1.17)
Competence satisfaction T2	**4.68 (1.20)**	**4.83_a (1.19)**	**4.79_a (1.17)**	**4.63_b (1.14)**	**4.72 (1.16)**
Competence frustration T1	4.21 (1.35)	4.11_b (1.38)	4.22 (1.34)	4.31_a (1.38)	4.25 (1.36)
Competence frustration T2	**4.04 (1.42)**	**4.00 (1.44)**	**4.00 (1.40)**	**4.08 (1.39)**	**4.03 (1.39)**

No tes: Overall: T1 $N = 1349$, T2 $N = 2090$; STEP: T1 $N = 497$, T2 $N = 591$; SI: T1 $N = 401$, T2 $N = 700$; Combined STEP & SI: T1 $N = 795$, T2 $N = 1124$; Comparison Units: T1 $N = 554$, T2 $N = 966$.
Significantly different scores based on t-tests ($p < 0.05$) at T2 than at T1 are shown in **bold.**
Within each row significant differences ($p < 0.05$) are shown with subscript where a > b.

at time-two than at time-one. Still, some effects of the intervention programmes were also suggested by the results. Students in the SI units experienced greater relatedness satisfaction and competence satisfaction at time-two than did students in the comparison units. Furthermore, students in units with the STEP interventions reported lower competence satisfaction at time-one but they 'caught up' and had similar competence satisfaction to students who were not part of the intervention programme by time-two. Finally, relatedness frustration was only significantly lower at time-two, on average, for students whose units were part of the STEP intervention. Taken together, these results suggest that the SI and STEP interventions had a positive impact on students' psychological need satisfaction and frustration as compared with the 'business-as-usual' comparison units. This broadly supports a possible hypothesis worth testing, that any targeted learning intervention will have a positive impact upon how related and satisfied the student feels, as a direct result of the intervention.

Retention and need satisfaction vs. frustration

In our CHAID analysis, we entered all demographic and student performance variables that were available to us from the university student records system in addition to NSFS scores. From this analysis, Grade Point Average (GPA) was the first-order significant predictor for re-enrolment/retention $\chi^2(7) = 3590.92$, $p < 0.001$. Specifically, students with higher GPAs were found to be more likely to enrol in units in the subsequent semester. Lower-order significant predictors of re-enrolment were mostly demographic factors such as age, disability, university entrance marks and gender. In addition, however, aspects of the intervention program and NSFS scores also predicted retention; specifically: time-two relatedness satisfaction ($\chi^2[1] = 18.30$, $p < 0.001$), attending SI sessions ($\chi^2[1] = 20.05$, $p < 0.001$), the number of SI sessions attended ($\chi^2[2] = 50.78$, $p < 0.001$) and enrolment in STEP units ($\chi^2[1] = 23.32$, $p < 0.001$). Overall, the CHAID analysis results indicated that a key reason for student re-enrolment was whether or not they passed their previous units. Beyond this, however, participation in the intervention units and satisfaction of students' psychological needs were related to increased progression.

Revenue modelling

Although there are clear benefits in terms of performance and retention for students, the related benefits for the institution were investigated as a way of quantifying these 'learning gains'. Around $200,000 was invested in the programme and the programme reached 2,990 students (65% of all first-year students). To extend our cost-benefit analysis, we configured a set of tables that showed the number of students who progressed to the next unit (module) against those who did not. We undertook this modelling by using the base unit (module fee) cost with the progression data. Ultimately, we used progression data to model the resulting benefit from the students undertaking the STEP and SI interventions.

This revenue modelling exercise found that the return-on-investment of $200,000 would be $486,320 in the first year just from the SI interventions. The other STEP interventions resulted in just under $1,000,000 return in the first year alone. The level of

Table 2. Cost benefit analysis of the SI programme.

Period	Units run	Students Involved	Student Retention*	Average Student Load	Est. Gross Revenue** (Per Semester)	Running Costs***	Revenue Estimate (Per Semester)
Sem 1 2015	4	305	24.4 (8%)	75.7%	$161,601	$63,237	$98,364
Sem 2 2015	5	286	28.6 (10%)	75.7%	$189,417	$79,046	$110,371
Sem 1 2016	13	759	61 (8%)	75.7%	$404,003	$94,985	$309,018
Sem 2 2016	15	555	42.2 (7.6%)	75.7%	$279,522	$102,188	$177,302
					$1,033,965	$339,456	$695,055

Notes: * The student retention estimate was determined by the amount of students who attended SI, re-enrolled the following semester in a load-bearing unit and persisted until census compared to students whom did not attend SI in the specified unit.

**Revenues only assume Domestic students ($17,500 per annum; excluding international students who attended; 10%, 2015 and 11%, 2016).

***Running costs are inclusive of all programme expenditure (including coordination salaries).

return from using interventions in 30 units is significant and has a 'multiplier-effect' each year over a 3- or 4-year cycle.

Overall, Table 2, which focuses on the SI programme as an example, shows substantial cost-benefits upon improving student-progression through a peer-learning intervention, and our experience of deploying SI was very positive with an average gain of 8% in student progression. Our cost-benefit analysis model of SI units (see Table 2 below) is based upon units run, student-progression, average student-load, estimated revenue and running costs. The cost-benefit analysis was only completed using the average revenue from domestic students (11% of the attendees in the program were international-students). A limitation of this analysis is the level of attribution of the programme to progression-gains. That is, we are wary about claiming 100% attribution to the SI intervention for student progression and therefore we suggest that this be modelled with an attribution analysis in the future.

In addition to modelling the financial-benefits, we were also interested in how we could relate progression improvements to interventions and possible revenue-recovery. Table 3 outlines the cumulative impact of improved retention by percentage for a single-year cohort of 4,822 students over 3 years (annual fees are $17,500). By linking the interventions, progression-rates and revenue-recovery, we can begin to plan future investment into proven and tested interventions that best support learning performance and progression improvements of the particular cohort. When considered along with studies into the use of student progression dashboard (e.g. de Freitas et al. 2015, 2017), it may be possible to track

Table 3. Revenue recoverable for each +1% increase in the retention rate (S:1 Y:2 to S2: Y:4).

%	1 Year	2 Years	3 Years
1	$843,850	$1,687,700	$2,531,550
2	$1,687,700	$3,375,400	$5,063,100
3	$2,531,550	$5,063,100	$7,594,650
4	$3,375,400	$6,750,800	$10,126,200
5	$4,219,250	$8,438,500	$12,657,750
6	$5,063,100	$10,126,200	$15,189,300
7	$5,906,950	$11,813,900	$17,720,850
8	$6,750,800	$13,501,600	$20,252,400

Note: For 4,822 students at $17,500.

student success and make changes and implement interventions 'on-the-fly', allowing for targeted and individual student adaptations

Discussion

Study findings

This study looked at the impact of a wide-scale innovative set of interventions applied within first-year university units at a single university. We approached the analysis from several different perspectives in order to gauge the impact of these interventions on engagement, progression, and, ultimately, the cost vs benefit of the interventions. Globally, we found that, on average, the interventions increased engagement as indicated by students' increased psychological need satisfaction and reduced psychological need frustration. Moreover, participation in units in which interventions were implemented, and in SI in particular, were predictive of increased student progression. Overall, these interventions cost less than the benefits gained from one semester of increased re-enrolments, and this benefit would compound if it translates to students progressing to subsequent semesters of study. Most importantly, the benefits to students themselves that may result from completing, rather than discontinuing, their education that may be attributed to support at first-year level may last a lifetime.

The initial results of this study are promising overall. Although it was not the purpose of the high-level analysis perspective we undertook to drill down to finer details, some of these are worth mentioning. For example, our study's findings suggest that, when one to three interventions are included in a unit (module), an uplift in student engagement leads to student-progression benefits. These interventions can be relatively simple and straightforward or more staff-intensive; e.g. we were pleasantly surprised to see that even quite minor STEP interventions (e.g. unit welcome-videos) had some impact on student engagement (via increased relatedness satisfaction). This corresponds with SDT predictions (Ryan and Deci 2000), and also with recent research findings that indicate the importance of strong staff–student relationships and a sense of belonging (O'Keeffe 2013; Pedler, Willis, and Nieuwoudt 2022; C. J. Williams, Dziurawiec, and Heritage 2018). More specifically, welcome-videos were short, low-cost, introductory videos that were provided by course-tutors for students. By providing initial induction information, which helped to form a connection between the student and the tutor, these videos would help to address SDT's core need of relatedness (especially for distance learners who were studying off-campus). Alongside these welcome-videos, the use of the writing-support software Grammarly was found to be quite beneficial for students (and may help to increase GPAs, which we found to be a key predictor of student progression). By comparison, although SI interventions proved to be extremely effective, these interventions represented a larger investment, and are more staff-intensive. Future studies will need to focus upon combinations of intervention types in different disciplines and undertake where possible pre- and post-surveys of students. A scale of intervention types, when compared among disciplines, could provide a useful tool for managers seeking to improve progression and enhance quality improvement.

Future studies should also focus upon assessment and its impact on student completion. In this light, researchers might expect even greater gains than seen in this study as

a result of reduced assessment and active-learning based curriculum-design enhancements. Indeed, as longitudinal studies of attrition have shown (de Freitas et al. 2015; C. Williams, de Freitas, and Dziurawiec 2018), failing assignments can lead to student dropout (especially during the first 1.5 years of study).

Reflections on the methodology

We recognise that the 'multi-layered methodology' outlined in this paper presents challenges, not least how to report this approach in a concise and comprehensible way. Because of this, the aim of this paper was to present a board-brush overview of the methods and broad details of the results rather than their minutia. We contend that this level of presentation is best for the purposes of informing higher education management decision-making regarding the efficacy and cost-benefit of learning interventions targeted at promoting progression. Moreover, in this paper we aimed to provide a model of how such interventions might be broadly evaluated. The strength of this approach is that, depending on the institution and context, numerous additional data sources and analyses can be 'slotted in', numerous hypotheses can be generated and models can be tested relatively easily.

Alongside the key results of this study, it is worth noting that the methodologies combined in this evaluation included a range of approaches. Ultimately, analytics such as those we collected could be fed into dashboards, which, if correctly designed, can provide immediate feedback to students, teaching staff and senior managers; and may be used to design effective and timely learning interventions (cf. Gibson and de Freitas 2016). In this way, analytics can offer a *dynamic*, rather than singularly *predictive*, set of data to augment the predictive models already used. Likewise, through such an approach, it is possible to target different interventions to certain students, according to the core needs (e.g. relatedness) that need to be addressed. This may allow university staff to create strong opportunities for reordering content and re-sequencing learning outcomes in a more adaptive way. In turn, this facilitates a reconsideration of standard testing/assessment methods (to reduce stress and dropout patterns).

Limitations

This study collected data in a quasi-experimental format, although we employed two times of testing for the NSFS, allocation of students to modules (units) was self-selected rather than random. Although such designs are the only ethical way to study such a large-scale intervention programme as reported in this study and provides a realistic representation of how such interventions typically occur, causal inferences concerning the impact of interventions on engagement and progression must be made cautiously. The very large sample size and scale of the study potentially help to overcome the design limitation as some sources of error attributable to non-random selection can be attenuated by sample size. Given that the university management chose the intervention units, and the research team chose the comparison units, it is not the case that students self-selected into interventions by choice.

In addition, as we have outlined above, the study sought to provide a broad-brush model of evaluation for a programme of the kind we studied rather than a fine detailed approach. Thus, although a range of interventions were included in the STEP program, it is beyond the scope of this paper to provide micro-level analysis of the impact and relative success of the various interventions. Because of this, as we mentioned earlier the overarching message from this paper is that a large set of interventions at first-year level can enhance engagement and increase progression, but more research is needed to understand which particular interventions are more efficient and effective.

Conclusion

In this study we investigated the implementation and outcomes of instituting a large number of educational, social, and technological interventions targeted at increasing student engagement and student progression. Given the sweeping nature of this intervention programme and the consequential lack of tight experimental controls that accompany such work, we aimed in this paper to present a multi-level and multi-faceted example of how such a programme can be evaluated. This evaluation can be used as a model for others who wish to undertake similar analyses of large-scale educational interventions at higher education institutions. In addition to this, we reported the outcome of this evaluation, which suggested that the programme we specifically evaluated increased student engagement, increased student progression and provided a positive return on investment.

The inherently multivariate nature of education makes single-method designs less useful in curricular decision-making and analysing the effectiveness of targeted interventions to achieve strategic objectives such as increased student progression. From our experience in this evaluation, effective evaluation designs should be multi-faceted and might need to be iteratively built-up over time (e.g. building models, benchmarks and runs of data that allow for patterns and discontinuities to be found and understood) to adequately capture the complexities and unique cultural attributes of each university.

Acknowledgement

The authors wish to thank contributors to this research, including Chris Cunningham, Frances Asha, Cameron Williams, and the unit coordinators, student advisors, and other staff and students who contributed to the PASS and STEP programs.

Disclosure statement

No potential conflict of interest was reported by the authors.

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Data availability statement

Raw data for the non-financial analyses reported in this paper are available from the corresponding author on request.

Ethics statement

Data collection methods were approved by the Murdoch University Human Research Ethics Committee (HREC 2016/240) and followed the protocols approved by that committee.

Consent statement

Consent to participate and consent to publish were obtained from all participants.

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